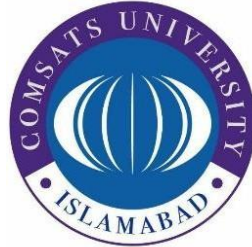


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INFORMATION SECURITY

Lab Project Report

Submitted To

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Secure Military Communication System

Using One-Time Pad (OTP) Stream Cipher

1. Introduction

In modern warfare and defense operations, secure communication is not merely a technical requirement it is a matter of national security and operational integrity. Military commands, troop movements, intelligence reports, and diplomatic messages must be transmitted without any risk of interception or decryption by adversaries. A single intercepted message can compromise an entire mission, endanger lives, or tip the balance of a conflict.

This project presents a Secure Military Communication System implemented using the One-Time Pad (OTP), a stream cipher that holds the rare distinction of being mathematically and provably unbreakable. Unlike modern symmetric ciphers such as AES or public-key systems like RSA, OTP achieves perfect secrecy not as an engineering goal, but as a mathematical guarantee proven by Claude Shannon in 1949.

2. Stream Cipher

A stream cipher is a type of symmetric encryption algorithm that encrypts data one bit or byte at a time, producing a keystream that is XOR-ed with the plaintext. Unlike block ciphers (which process fixed-size chunks), stream ciphers are inherently suited to real-time, continuous data streams making them ideal for communications, satellite transmissions, and embedded military hardware.

3. One-Time Pad (OTP)

The One-Time Pad is the simplest and most secure form of stream cipher. It is the only known cipher offering information-theoretic security, meaning it cannot be broken even with unlimited computational resources.

For OTP to guarantee perfect secrecy, four conditions must be strictly met:

- The key must be truly random (not pseudo-random).
- The key must be at least as long as the plaintext.
- The key must never be reused not even partially.
- The key must be kept completely secret from all unauthorized parties.

4. Selected Encryption Technique

World War II Intelligence: Both Allied and Axis powers used one-time pad systems for top-secret communications; the Soviet VENONA project used flawed OTP (key reuse), which eventually allowed partial American decryption.

For this project, the selected encryption technique is the One-Time Pad (OTP) Stream Cipher. OTP was chosen over other modern ciphers such as AES-256, RSA, and ChaCha20 due to its unique theoretical property of perfect secrecy, which makes it the most suitable choice for a military-grade communication system where no compromise on security is acceptable.

Classification

OTP is classified as follows in the taxonomy of cryptographic algorithms:

- Type: Symmetric-key encryption
- Category: Stream cipher
- Security model: Information-theoretic (unconditional) security
- Core operation: Bitwise XOR (Exclusive OR)

5. Why OTP Over Other Ciphers?

Property	OTP	AES-256	RSA-2048	ChaCha20
Perfect Secrecy	Yes	No	No	No
Quantum Resistant	Yes	Partial	No	Partial
Speed	Very Fast	Fast	Very Slow	Fast
Key Reuse	Never	Allowed	Allowed	Allowed
Scalability	Low	High	High	High
Complexity	Simple	Moderate	Complex	Moderate

6. Python Implementation

IMPLEMENTATION

```

Project.py > ...
1  # GROUP MEMBERS: RAFIA RAMEEN & SAMRINA MANAHIL
2  import os
3  import sys
4
5  # OTP Key Generation
6  def generate_key(length):
7      return list(os.urandom(length))
8
9  # OTP Encryption using XOR
10 def encrypt(plaintext, key):
11     return [p ^ k for p, k in zip(plaintext, key)]
12
13 # OTP Decryption
14 def decrypt(ciphertext, key):
15     return encrypt(ciphertext, key)
16
17 # Main Secure Communication Demo
18 def secure_military_comm():
19     print("=" * 50)
20     print("    SECURE MILITARY COMMUNICATION SYSTEM")
21     print("    One-Time Pad Encryption")
22     print("=" * 50)
23
24     message = input("Enter Message: ")
25     plaintext = list(message.encode())
26
27     key = generate_key(len(plaintext))
28     ciphertext = encrypt(plaintext, key)
29     recovered = decrypt(ciphertext, key)
30
31     print("\n--- Communication Results ---")
32     print("Original Message :", message)
33     print("Encrypted Data    :", ciphertext)
34     print("Decrypted Message:", bytes(recovered).decode())
35
36     secure_military_comm()

```

7. Output Sample

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + v [ ] [ ] ... [ ]

PS D:\BSE-4 RAFIA\Information Security\IS-LAB\IS-LAB-PROJECT> & "C:\Users\Discount Laptop\AppData\Local\Programs\Python\Python314\python.exe" "d:/BSE-4 RAFIA/Information Security/IS
AB/IS-LAB-PROJECT/Project.py"
=====
    SECURE MILITARY COMMUNICATION SYSTEM
    One-Time Pad Encryption
=====
Enter Message: CYBER ATTACK ON POWER GRID CONFIRMED

--- Communication Results ---
Original Message : CYBER ATTACK ON POWER GRID CONFIRMED
Encrypted Data   : [168, 6, 157, 25, 35, 65, 122, 7, 9, 115, 222, 56, 56, 48, 175, 89, 117, 95, 42, 30, 212, 51, 199, 138, 162, 231, 232, 255, 223, 255, 224, 224, 47, 165, 0, 241]
Decrypted Message: CYBER ATTACK ON POWER GRID CONFIRMED

```

8. Advantages

- **Perfect Secrecy (Information-Theoretic Security):** Proven by Claude Shannon (1949), OTP is the only cipher that is mathematically unbreakable regardless of the attacker's computational power or time. No pattern can be extracted from the ciphertext alone.
- **Quantum Computer Resistant:** Unlike AES or RSA, OTP is immune to quantum attacks (such as Grover's or Shor's algorithms) because its security does not rely on computational hardness it relies on information theory.
- **Simplicity of Operation:** The XOR operation is among the fastest operations a CPU can perform (single clock cycle). Encryption and decryption are both $O(n)$ and require no complex arithmetic, making OTP extremely efficient in constrained environments.
- **No Ciphertext-Only Attack Possible:** Without the key, an attacker sees only random noise. Every possible plaintext of that length is equally likely an attacker gains zero information from the ciphertext.
- **Ideal for High-Security Military Use:** Used in the US-USSR hotline, battlefield command systems, and intelligence agencies precisely because its security does not degrade with computational advances.

9. Disadvantages

- **Key Length Equals Message Length:** For a 1 MB message, a 1 MB key is required. For high-volume military transmissions, this becomes logistically extreme.
- **Strict Single-Use Requirement:** If any portion of the key is reused for a different message, the cipher is catastrophically broken. XOR-ing two ciphertexts encrypted with the same key reveals the XOR of the two plaintexts, which can be linguistically analyzed (this is what defeated the Soviet VENONA program).
- **Key Distribution Problem:** Transmitting the key securely to the receiver is as hard as transmitting the message itself. In practice, OTP keys are delivered physically (e.g., military couriers, secure diplomatic pouches).
- **Not Scalable for Modern Internet:** The internet requires billions of unique encrypted connections daily. OTP's per-message key requirement makes it completely impractical at internet scale.

10. Security Analysis

- **Shannon's Perfect Secrecy**

Proven by Claude Shannon in 1949

OTP is the only mathematically unbreakable cipher in existence

Our system satisfies all 4 conditions: random key, key length = message length, never reused, kept secret.

- **Brute-Force Attack**

Attacker tries every possible key combination

Each key produces a different valid-looking plaintext no way to identify the correct one

Result: Completely Immune

- **Cipher-Text Only Attack**

Attacker intercepts the ciphertext and tries to decode it

OTP ciphertext looks like pure random noise no patterns, no structure

Result: Completely Immune